



GALGOTIAS UNIVERSITY
SCHOOL OF ENGINEERING & TECHNOLOGY (SET)

Department of Computer Science & Engineering • Software Engineering Laboratory

FORMAL LABORATORY MANUAL & MODULAR OUTCOMES REPORT

PROJECT: LOANZO

Institutional Peer-to-Peer (P2P) Lending & Credit Ledger Ecosystem

Core Software Engineering Principle — Independent Modular Outcomes:

A software component is strictly classified as an independent **Module** if and only if it exhibits *high cohesion*, *loose coupling*, and delivers a **unique, mutually exclusive, verifiable outcome artifact** that no other module produces.

- **Lab 2 Outcome:** Problem Definition Document & Modular Scope Analysis
- **Lab 3:** Feasibility Study (Technical, Economic, Operational Feasibility & Constraints)
- **Lab 4:** Requirement Elicitation – I (Stakeholder Identification, Methodologies, FR, NFR & MoSCoW)
- **Master Traceability:** End-to-End Requirements Traceability Matrix linking all 7 Modules with their Unique Outcomes.

Academic Term:	Fall Semester 2026	Course Code:	BCSE308P / SE Lab
Subject:	Software Engineering	Modularity Standard:	IEEE 1016 & ISO/IEC 25010
Target Platform:	Android 8.0+ (Jetpack Compose)	Security Standard:	RBI P2P NBFC & IT Act 2000 Sec 10A
Author / Team:	Student Engineering Team	Evaluation Date:	September 2026

Executive Overview: The 7 Core System Modules & Their Unique Outcomes

Under modular software architecture, a system decomposition is valid only if each module possesses a distinct functional boundary and produces a **unique, verifiable outcome artifact**. Below is the formal specification of Loanzo's 7 architectural modules and their exclusive deliverables:

Module #	Module Name	Unique Verifiable Outcome Artifact (Exclusive Deliverable)	Core Responsibilities & Capabilities
M-01	Auth, Biometrics & Session Guard	HardwareBoundSessionToken & CryptographicSecurityClearance (Hardware UID `UID-XXXXXXXXXX`, In-Memory Pre-Warmed Cache, Active/Locked Session State)	Multi-factor authentication, AndroidX Biometrics, hardware device binding, 3.5s parallel startup warm-up, 3m auto-lock & 2d hard session expiry.
M-02	KYC & Encrypted Document Vault	VerifiedKYCDossier & DecryptedVaultRecord (Government DigiLocker Token, Liveness-Verified Face Embedding, AES-256 Decrypted ID Cards)	DigiLocker OAuth (Aadhaar/PAN), ML Kit Face Liveness Detection, AES-256 encrypted vault with password/biometric gating.
M-03	Bilateral Loan Origination & Matching	OriginatedLoanProposal & AmortizationSchedule (Bilateral Loan Contract terms, Counterparty Binding, EMI Math Curve)	Unified Consumer Dashboard (Lender & Borrower perspectives), Grant/Request Loan flows, multi-attribute counterparty binding, interactive Loan Simulator.
M-04	Multi-Tranche & Repayment Ledger	ImmutableAuditLedger & TrancheSettlementProof (Room SQLite v15 Synchronized Transaction History, Outstanding Balance, Milestone Releases)	Milestone-based tranche disbursement, micro-repayment amortization tracking, overdue penalty engine, Room SQLite v15 offline-first sync.
M-05	Legal Contract & Dual e-Sign Engine	LegallyBindingPromissoryPDF & DigitalSignatureHash (Tamper-evident contract PDF, Touch eSign, Liveness Stamp, IT Act 2000 Sec 10A SHA-256 Hash)	On-device OpenPDF agreement generator with official app logo, canvas-based signature capture, IT Act 2000 Sec 10A digital validity.
M-06	Empaneled Field Agent Inspection	GeoTaggedCollateralAppraisalReport (GPS-Stamped Physical Asset Record, Lat/Long Coordinates, Tamper-Evident Photo Proof, Duty Payout Voucher)	Doorstep collateral verification, GPS geotagged photo evidence, duty status toggling, role-isolated agent cockpit.
M-07	Alerts, Telegram Bot & Localization	DisputeDocket & MultiChannelAlertDispatch (Formal Dispute Docket, Automated 3-Day/1-Day Overdue Repayment Reminders, Telegram Webhook Notice)	Automated 3-day/1-day/today deadline scanner, formal dispute arbitration center, 24/7 Telegram Assistant Bot, 11-language Devnagari localization.

Part I: Problem Definition Document (Lab 2 Outcome)

1.1 Background & Macro-Economic Problem Context

India's informal credit market exceeds **₹15 Lakh Crore (\$180 Billion USD)** annually, serving micro-borrowers, rural artisans, and small traders neglected by institutional commercial banks. Despite high transaction velocity, the sector relies entirely on unorganized mental accounting, paper diaries (Bahi-Khata), and informal verbal agreements. When default, death, or dispute occurs, participants have zero legal standing, leading to total capital loss, financial ruin, or predatory harassment.

1.2 Modular Problem Mapping & Unique Problem-Solution Pairs

Module	Existing Industry Deficiency / Root Problem	Unique Outcome Artifact That Solves It
M-01: Auth & Session	Fintech apps either constantly log users out (frustrating OTP entry) or leave sensitive balances open to shoulder-surfing when checking SMS.	HardwareBoundSessionToken: 3-min background grace window for copying OTPs; auto-lock to biometric shield after 3m; 2-day hard expiry.
M-02: KYC & Vault	Forged identity documents, unverified PAN cards, and sensitive Aadhaar copies exposed unencrypted in device photo galleries.	VerifiedKYCDossier: DigiLocker direct government verification, ML Kit face liveness detection, and AES-256 encrypted document vault.
M-03: P2P Matching	Borrowers and lenders operate in disconnected silos; conventional apps force users to register as either a lender OR a borrower, never both.	OriginatedLoanProposal: Unified Consumer Scaffold enabling dual-role participation, peer discovery, bilateral binding, and interactive loan amortization calculator.
M-04: Repayment Ledger	Lump-sum informal loans lead to high default; parties dispute whether payments covered interest or principal; rural offline outages break apps.	ImmutableAuditLedger: Multi-tranche milestone disbursement, transparent interest/principal breakdown, and Room SQLite offline-first database with auto-sync.
M-05: Legal Contracts	Informal promissory notes are verbal or drawn on plain paper, rendering them legally inadmissible under IT Act 2000 Section 10A.	LegallyBindingPromissoryPDF: On-device OpenPDF engine generates binding multi-page contracts with Aadhaar e-Sign, liveness photo embed, and cryptographic SHA-256 hash.
M-06: Field Inspection	Lenders cannot physically inspect distant collateral (vehicles, gold, machinery) or verify borrower residence, leading to ghost assets.	GeoTaggedCollateralAppraisalReport: Empaneled Doorstep Field Agent network with role isolation, GPS geotagging, inspection checklists, and tamper-evident photo capture.
M-07: Alerts & Support	Missed repayments turn into bad debts due to lack of reminders; legal disputes have no fast mediation; non-English users excluded.	DisputeDocket & MultiChannelAlertDispatch: Automated 3-day/1-day deadline scanner, formal dispute arbitration center, 24/7 Telegram Assistant Bot, and 11-language Devnagari UI.

Part II: Lab 3 — Comprehensive Feasibility Study Report

2.1 Technical Feasibility Analysis across All 7 Modules

Technical feasibility validates whether the required engineering stacks, computational complexity, memory limits, and networking constraints can support production scale.

System Module	Core Technologies & APIs	Technical Feasibility & Engineering Justification
M-01: Auth & Security	AndroidX Biometric 1.2.0, DataStore, SHA-256 UID	Hardware-backed Keystore, zero biometric credential leakage to cloud, 3.5s parallel startup pipeline eliminates UI layout jumps completely.
M-02: KYC & Vault	DigiLocker OAuth 2.0, ML Kit Face, SQLCipher	Government verified Aadhaar/PAN validation; on-device facial landmark liveness prevents 2D photo spoofing without network latency.
M-03: Loan Matching	Jetpack Compose M3, Coroutines, StateFlow	Unidirectional Data Flow (UDF) guarantees single source of truth across bilateral tabs with sub-16ms Compose frame rendering.
M-04: Repayment Ledger	Room SQLite v15, WorkManager SyncWorker	Guarantees 100% offline readability. Offline tranche updates queue in SyncQueueEntity and automatically flush to Firestore upon reconnection.
M-05: Legal Contracts	OpenPDF / Android Canvas, Google Apps Script Proxy	Client-side PDF compilation completes in < 450ms. Bypasses expensive cloud doc APIs; stores signed hashes securely on Google Drive.
M-06: Field Inspection	FusedLocationProvider, CameraX, GeoStamper	GPS accuracy within 5 meters; cryptographic EXIF metadata stamping prevents location spoofing by inspection officers.
M-07: Alerts & Telegram	Telegram Bot API (OkHttp), Android Configuration	Async webhook dispatch with retry backoff; dynamic locale configuration switcher updates Devnagari strings without restarting Activity.

2.2 Economic Feasibility Analysis (Cost-Benefit & Serverless Economics)

Economic feasibility proves that Loanzo can operate with near-zero marginal infrastructure costs by offloading computation to client devices and leveraging public digital infrastructure.

Cost Element	Traditional Cloud Fintech	Loanzo Client-Heavy Model	Financial Impact
Compute / CPU	■60,000 / mo (Cloud VM cluster)	■0 / mo (On-device Android CPU)	100% cost reduction for PDF & KYC rendering.
Database Reads/Writes	■35,000 / mo (High IOPS database)	■0 / mo (Local Room SQLite Cache)	90% query traffic resolved locally on device.
E-Sign Licensing	■12 per signature (DocuSign)	■0 (Native OpenPDF canvas engine)	Zero ongoing operational licensing fees.
Third-Party SMS OTP	■0.25 per SMS (High monthly cost)	DigiLocker OAuth + Telegram Alerts	80% reduction in transactional telecom bills.

2.3 Operational Feasibility & Regulatory Alignment

- **User Operational Adoption:** Intuitive UI modeled after Cred and Revolut featuring an Executive Drill-down Hub, large touch targets, and full Devnagari bilingual support for Tier-2/3 Indian cities.
- **RBI NBFC-P2P Compliance:** Adheres strictly to Reserve Bank of India Master Directions on P2P Lending (2017): Loanzo acts as a technology intermediary; capital transfers occur bank-to-bank via UPI/IMPS; no retail deposits held.
- **Legal Admissibility (IT Act 2000):** Section 10A explicitly recognizes electronic contracts. Loanzo agreements include party timestamps, IP logs, Aadhaar references, and SHA-256 tamper hashes admissible in Indian courts.

2.4 Constraints and Assumptions Matrix

Technical Constraints: (1) Android SDK minimum version 26 (Android 8.0 Oreo); (2) Device biometric sensor required for quick unlock; (3) Offline cache constrained by available device flash storage.

Regulatory Constraints: (1) Maximum aggregate P2P lending cap of ■50,00,000 across all platforms per lender; (2) Data localization mandated within Indian sovereign jurisdiction.

Key Project Assumptions: (1) Counterparties possess basic smartphone familiarity; (2) At least one counterparty has active internet access periodically to sync the bilateral ledger.

Part III: Lab 4 — Requirement Elicitation – I

3.1 Comprehensive Stakeholder Identification & Salience Model

Stakeholder identification mapped all individuals, groups, and statutory entities impacted by the platform. A 4-tier salience model was developed:

Stakeholder Group	Salience Tier	Primary Business Needs	System Touchpoints & Interfaces
Retail Borrower (Student / Micro-Merchant)	Primary (Core User)	Fair interest rates, transparent tranches, zero predatory harassment, emergency loans.	Request Loan flow, Repayment tracking, DigiLocker KYC, Document Vault.
Peer Lender (Salaried / Angel / NRI)	Primary (Core User)	High risk-adjusted yield, legally enforceable contracts, verified collateral, default alerts.	Grant Loan flow, Legal PDF contract, Collateral inspection report, Loan Ledger.
Certified Field Agent	Secondary (Field Officer)	Clear inspection queues, offline inspection tools, fair per-visit compensation, safety.	Agent Cockpit, Collateral Inspection Sheet, Duty status toggle, GPS camera.
Co-Signer / Guarantor	Secondary (Risk Mitigator)	Transparent liability disclosure, automated notifications when borrower defaults.	Guarantor commitment screen, Digital signature flow, Alert center.
Platform Master Admin	Tertiary (Governance)	Fraud prevention, dispute resolution, agent empanelment, system health audit.	Master Admin Hub, Dispute Action Sheet, Telegram Admin alerts.
Regulatory Auditor (RBI)	Tertiary (Compliance)	Verification of lending limits, digital contract compliance, zero deposit holding.	Immutable Room audit trail, SHA-256 agreement verification hashes.

3.2 Requirement Elicitation Methodologies Executed

- Structured & Semi-Structured Interviews:** Engaged 45 micro-merchants and 25 retail lenders to uncover informal credit trust patterns.
- Bahi-Khata & Chit-Fund Artifact Analysis:** Reviewed real-world physical promissory notes and ledger entries to model bilateral loan states.
- Interactive Jetpack Compose Prototype Walkthroughs:** Evaluated usability across 30 smartphone users to calibrate the 3-minute grace window and 3.5s logo splash animation.

3.3 Functional Requirements (FR) Matrix Across All 7 Modules

Req ID	Module	Functional Requirement Description	Priority
FR-01	M-01: Auth	System shall enforce hardware device fingerprint binding (UID-XXXXXXXXXXXX) on login.	MUST
FR-02	M-01: Auth	System shall auto-lock to Biometric Shield after 3 minutes in background; hard logout after 2 days.	MUST
FR-03	M-02: KYC	System shall authenticate identity via DigiLocker OAuth and ML Kit facial liveness detection.	MUST
FR-04	M-02: KYC	System shall protect PAN/Aadhaar in an AES-256 encrypted vault requiring password/fingerprint.	MUST
FR-05	M-03: Loan	System shall provide unified bilateral dashboard supporting Grant Loan and Request Loan.	MUST
FR-06	M-03: Loan	System shall provide an interactive Loan Simulator computing monthly EMI and amortization.	SHOULD
FR-07	M-04: Ledger	System shall record multi-tranche disbursements and compute outstanding balance offline.	MUST
FR-08	M-05: Contract	System shall compile a multi-page legal agreement PDF on-device with official app logo & eSign.	MUST
FR-09	M-06: Agent	System shall isolate certified agent cockpit and record GPS geotagged inspection evidence.	SHOULD
FR-10	M-07: Alerts	System shall scan active loans daily and generate 3-day, 1-day, and today deadline notifications.	MUST
FR-11	M-07: Alerts	System shall link user account to Telegram Bot (@Loanzo_bot) for instant administrative alerts.	COULD

3.4 Non-Functional Requirements (NFR) Specification

- NFR-01 (Startup Warm-Up Performance):** Cold startup must execute Room prefetching and device binding within the 3.5s splash duration, guaranteeing Oms screen jump when rendering the dashboard.
- NFR-02 (Session Security Invariant):** Minimizing app for < 180s must resume without re-auth. Backgrounding for >= 180s must lock into biometric shield. Inactivity > 48 hours must trigger hard logout.
- NFR-03 (Offline Availability):** 100% of existing loan contracts, repayment histories, and document vault items must be accessible without an active internet connection.
- NFR-04 (Cryptographic Data Integrity):** All loan agreements and collateral reports must have a SHA-256 verification hash computed on generation.

Part IV: Master End-to-End Requirements Traceability Matrix (RTM)

The ****Master Requirements Traceability Matrix (RTM)**** establishes rigorous bidirectional traceability across all software engineering phases: from Problem Definition (Lab 2), through Feasibility Analysis (Lab 3), to Stakeholder Elicitation (Lab 4), highlighting the ****Unique Verifiable Outcome Artifact**** that validates each component as an independent module.

Mod ID	Module Name	Unique Outcome Artifact (Exclusive Deliverable)	Lab 2 Problem Addressed	Lab 3 Tech Feasibility	Lab 4 Stakeholder	Mapped FR/NFR & MoSCoW
M-01	Auth & Session Security	HardwareBoundSessionToken & SecurityClearance	Credential theft, APK cloning & abrupt screen jumps.	AndroidX Biometric, SHA-256 UID, 3.5s async warm-up.	All Users (Borrower, Lender, Agent)	FR-01, FR-02 NFR-01, NFR-02 [MUST]
M-02	KYC & Document Vault	VerifiedKYCDossier & DecryptedVaultRecord	Fake identity cards, unencrypted documents in gallery.	DigiLocker OAuth, ML Kit Liveness, AES-256 Room.	Retail Borrower, Peer Lender, RBI Auditor	FR-03, FR-04 NFR-03, NFR-04 [MUST]
M-03	Bilateral Loan Origination	OriginatedLoanProposal & AmortizationCurve	Siloed borrower/lender apps, opaque interest terms.	Jetpack Compose UDF architecture, sub-16ms simulator.	Retail Borrower, Peer Lender	FR-05, FR-06 NFR-01, NFR-05 [MUST]
M-04	Repayment & Tranche Ledger	ImmutableAuditLedger & SettlementProof	Lump-sum default risk, disputed interest/principal.	Room SQLite v15 offline persistence, WorkManager.	Retail Borrower, Peer Lender, Platform Admin	FR-07 NFR-03, NFR-04 [MUST]
M-05	Legal Contract & e-Sign	LegallyBindingPromissoryPDF & SignatureHash	Verbal or plain-paper notes inadmissible in court.	Client-side OpenPDF, canvas touch signature, zero cost.	Borrower, Lender, Guarantor, Arbitrator	FR-08 NFR-03, NFR-04 [MUST]
M-06	Field Agent Inspection	GeoTaggedCollateralAppraisalReport	Ghost collateral, unverified borrowers, inspection fraud.	FusedLocationProvider GPS, CameraX EXIF proof.	Certified Field Agent, Peer Lender, Master Admin	FR-09 NFR-04 [SHOULD]
M-07	Alerts, Disputes & Bot	DisputeDocket & MultiChannelAlertDispatch	Missed repayment default, non-English exclusion.	Telegram Bot API webhook, daily deadline scanner.	All Users, Master Admin, Regulatory Auditor	FR-10, FR-11 NFR-05 [SHOULD/COULD]

4.2 Stakeholder-to-Module Cross-Reference Matrix

Stakeholder Persona	M-01 Auth	M-02 KYC	M-03 Loan	M-04 Ledger	M-05 Legal	M-06 Agent	M-07 Alerts
Retail Borrower	✓ Full	✓ Full	✓ Full	✓ Full	✓ eSign	— Read	✓ Alerts
Peer Lender	✓ Full	✓ Full	✓ Full	✓ Full	✓ eSign	✓ Order	✓ Alerts
Certified Field Agent	✓ Full	— Verify	— None	— None	— None	✓ Cockpit	✓ Payouts
Guarantor / Co-Signer	✓ Auth	✓ KYC	— None	— Read	✓ Co-Sign	— None	✓ Default
Platform Master Admin	✓ Master	✓ Override	✓ Monitor	✓ Audit	✓ Archive	✓ Empanel	✓ Telegram
RBI / Legal Auditor	— Audit	✓ Audit	— Caps	✓ Ledger	✓ SHA-256	✓ Geotag	✓ Reports

4.3 Lifecycle Progression Mapping (From Problem to Verification)

Phase 1 (Problem Definition): Identifies informal credit systemic failures (usurious compounding rates, untracked repayments, lack of legal proof, and APK cloning).

Phase 2 (Feasibility Study): Benchmarks client-heavy architecture (Room SQLite v15, AndroidX Biometrics, on-device OpenPDF engine) delivering **99% infrastructure cost savings** with 100% offline availability and RBI P2P NBFC compliance.

Phase 3 (Requirement Elicitation): Elicits and maps functional rules (FR-01 to FR-11) and security invariants (NFR-01 to NFR-05) across all 4 stakeholder salience tiers.

Phase 4 (Production Implementation & Verification): Compiles into production Android APK with verified unit tests, 3.5s parallel splash warmup, and banking-grade session auto-lock.

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Laboratory Evaluation & Viva Defense Sign-Off:

This comprehensive document provides complete, bidirectional traceability for **Lab 2 (Problem Definition)**, **Lab 3 (Feasibility Study)**, and **Lab 4 (Requirement Elicitation – I)** across all 7 production modules of the Loanzo P2P Lending System, strictly verifying that each module produces a unique, exclusive outcome artifact.

Verified by Faculty Evaluator: _____ **Marks Awarded:** _____ / 30